Comparative Epidemiological Study — Tocantins (TO), Amapá (AP) and Rondônia (RO)

Cardiovascular/Thromboembolic and Oncology Therapeutic Areas — ShokAI Project (Daiichi Sankyo)

Business Understanding / Data Understanding Phase — CRISP-DM Methodology

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Methodological Note

This document was prepared from assisted desk research (web-research agents) conducted on August 24, 2026, prioritizing primary Brazilian sources (DATASUS/SIM, SIH, SINAN, CNES; IBGE; INCA; ANS; Ministério da Saúde; state health secretariats SESA/SESAU) and academic literature (SciELO, PubMed, Cadernos de Saúde Pública, Revista de Saúde Pública, Ciência & Saúde Coletiva). Citations follow Vancouver style and are numbered in order of first appearance in the text. Reference titles are kept in their original language of publication (mostly Portuguese), as is standard academic practice; only the surrounding narrative is in English.

Limitations stated explicitly, for academic transparency:

1. **Tocantins**: dedicated desk research for this state was interrupted before completion, on the user's explicit instruction. The Tocantins section below therefore contains only data found incidentally during the cross-cutting Brazil/North Region research; for the overwhelming majority of the requested points, the label **"data not collected"** is used. This does not mean the data does not exist — only that it was not researched in this session.
2. **Amapá and Rondônia**: research was substantially more thorough, but several interactive interfaces (DATASUS TabNet for cause-of-death detail by ICD-10 chapter at state level; CNES registry by individual medical specialty) could not be queried to completion in this session; these gaps are flagged at each occurrence with **"data not collected."**
3. Every figure is accompanied by its reference date and source. No figure is extrapolated or estimated without this being explicitly flagged.
4. Vigitel surveys, by design, cover only state capitals (Palmas, Macapá, Porto Velho) — never a representative sample of the whole state; this is restated at every use.

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1. Executive Summary

Brazil recorded **1,542,157 deaths in 2022** (DATASUS/SIM), of which diseases of the circulatory system (**397,920**, 25.8%) and neoplasms (**242,963**, 15.8%) were respectively the 1st and 2nd leading causes of death, followed by respiratory diseases (176,111), external causes (152,090),

and infectious and parasitic diseases (131,856) [1]. The National Cancer Institute (INCA) estimates **704,000 new cancer cases per year** in Brazil for the 2023-2025 triennium [2] [3], and national health-plan (ANS) coverage reached **25.0% of the population as of June 2026** (53.1 million beneficiaries) [4].

The three states studied — Tocantins, Amapá, Rondônia — share one major, well-documented feature for two of them: supplementary health coverage far below the national average (Amapá 7.8%, Rondônia 9.1%, Tocantins 7.8%, versus 24.2% for Brazil as of December 2024) [5], implying near-total dependence on the public SUS system for the vast majority of the population in all three states — a structural fact for interpreting any analysis built on ANS/TISS microdata.

For **Amapá** and **Rondônia**, research produced a reasonably detailed picture: the cardiovascular burden (mortality from stroke/myocardial infarction, heart failure, SIH hospitalizations), INCA cancer-incidence estimates, several environmental determinants specific to the Amazon (mercury exposure linked to illegal mining, possible effects of biomass-burning smoke), the persistence of infectious diseases (malaria, dengue, tuberculosis), the racial/ethnic profile of the population (a high share of people self-identifying as "pardo," fast-growing Indigenous populations), and the structure of the care supply (CNES, TFD — out-of-domicile treatment referrals). For **Tocantins**, by contrast, most of the seven guiding questions remain undocumented in this session; only a few isolated data points (2023 overall mortality, 2023 SIH hospitalizations, ANS coverage rate) were retrieved incidentally during the cross-cutting Brazil/North Region research.

The report closes with three hypotheses directly testable on ANS/TISS microdata (available columns: operator, provider, procedure code, event value, age, sex, municipality), each linking a documented determinant to an expected "patient potential."

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2. Tocantins (TO)

Caveat: this section is deliberately incomplete. Dedicated desk research on Tocantins was interrupted before completion on the user's instruction. Only the data below, obtained incidentally during the comparative Brazil/North Region research, are available. For every other point, "data not collected" applies.

2.1 Leading Causes of Morbidity and Mortality

According to a TabNet extraction from the Mortality Information System (SIM/DATASUS), Tocantins, year 2023 [6]:

Cause (ICD-10 chapter)	Deaths (2023)	Share of total
Total, all causes	**8,912**	100%
Diseases of the circulatory system	**2,506**	28.1%
External causes	**1,508**	16.9%
Neoplasms (tumors)	**1,377**	15.5%

This profile (circulatory diseases ranked 1st, neoplasms 3rd, behind external causes) differs slightly from the national structure, where neoplasms rank 2nd (15.8% nationally versus 16.9% for external causes in Tocantins) [1] [6] — a full quantified comparison (age-standardized rates, time trends) against the national and North Region averages **could not be established** in this session: **data not collected.**

2.2 Cardiovascular and Thromboembolic Disease Burden

Only available data point: **6,917 SIH/SUS hospitalizations** for diseases of the circulatory system (ICD-10 chapter IX) recorded in Tocantins in 2023, versus 80,380 for the North Region as a whole and 1,279,882 for Brazil [7]. Incidence, prevalence, and mortality by specific condition (stroke, myocardial infarction, heart failure, atrial fibrillation, deep vein thrombosis, pulmonary embolism): **data not collected.**

2.3 Cancer Burden

INCA incidence estimates by cancer type, mortality, and stage at diagnosis: **data not collected** for Tocantins in this session (unlike Amapá and Rondônia, for which dedicated INCA state pages were consulted).

2.4 Determinants (Epidemiological Transition, Behavioral Factors, Migration, Environment, Infectious Diseases)

Data not collected. No targeted research was conducted on Tocantins' demographic transition, Vigitel data for Palmas, migration patterns specific to this state (created in 1988, the most recent of Brazil's 27 federative units), or the persistence of malaria, dengue, or tuberculosis on its territory.

2.5 Population Profiles Affected

Data not collected: age/sex structure, racial/color composition, Indigenous population (Tocantins hosts peoples such as the Xerente and Krahô, but no quantified demographic or health data on them was gathered), income, education, rural vs. urban distribution.

2.6 Healthcare Supply Structure

Only available data point: the health-plan (ANS) coverage rate, **7.8% as of December 2024**, identical to Amapá's and far below the national average (24.2% at the same date) [5]. Tocantins runs a dedicated portal for Tratamento Fora de Domicílio (TFD, out-of-domicile treatment referral, central3.to.gov.br), confirming that this referral mechanism exists structurally for patients requiring procedures unavailable locally [8], but **no patient volumes, no preferred destinations (Palmas, Goiânia, Brasília), and no figures on physician density, CNES facility counts, or cardiologist/oncologist counts were gathered: data not collected.**

2.7 Access Inequalities

Data not collected.

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3. Amapá (AP)

3.1 Leading Causes of Morbidity and Mortality

The state's strategic plan against chronic disease (SESA/SVS-AP, 2024) identifies non-communicable chronic diseases (cardiovascular, cancer, diabetes, chronic respiratory) as the leading cause of morbidity and premature mortality in the state [9]. The premature-mortality rate (ages 30-69) from these four conditions rose from **214.1 per 100,000 in 2018 to 252.0 in 2022** (+18% between 2019 and 2022), with a dip in 2021 (221.0, likely a pandemic effect on death-cause reporting/prioritization) [9]. The Central health micro-region (Macapá/Santana) shows the highest rate (234 in 2022) [9].

A direct, dated, quantified comparison with the national average and the North Region for this composite indicator: data not collected — only disease-by-disease comparisons could be documented (see 3.2 and 3.6). The burden of infectious disease remains high, particularly malaria (see 3.4).

3.2 Cardiovascular and Thromboembolic Disease Burden

Stroke and myocardial infarction — retrospective study on SIM/DATASUS data, 2015-2024 [10]: **1,224 deaths** from stroke (ICD-10 I64) and **1,911 deaths** from acute myocardial infarction (ICD-10 I21) in total over the period; the highest premature-mortality rate observed was for stroke in 2023 (**18.1 per 100,000**) and for myocardial infarction in 2024 (**41.8 per 100,000**).

Heart failure — five-year trend study (SIH/SUS), 2019-2024 [11]: in-hospital case-fatality rate of **14.5%**; profile of hospitalized patients: 63% male, **82.6% self-identifying as "pardo,"** 68.6% aged over 60, 75.8% of cases concentrated in the Macapá/Santana metropolitan region; overall trend **stable** over the period, but a rising burden among older men.

Diabetes (comorbidity/risk factor): **16,830 patients** followed in primary care (e-SUS-AB) in 2024 [9].

Atrial fibrillation, deep vein thrombosis, pulmonary embolism: data not collected — no Amapá-specific study on these three conditions was identified; they are likely folded into the generic "diseases of the circulatory system" statistics in SIM/SIH without a dedicated breakdown available. A dashboard on cardiovascular mortality in the Legal Amazon maintained by the National Institute of Cardiology exists, but its detailed figures could not be extracted in this session [12].

3.3 Cancer Burden

According to the most recent available INCA estimate (Amapá state page, updated February 4, 2026, *Estimativa 2023* methodology) [13] [3]:

Indicator	Value
Total estimated new cases/year (excluding non-melanoma skin cancer)	**1,140**
Men	510 (crude rate 126.00 per 100,000)
Women	630 (crude rate 155.49 per 100,000)

Most frequent types: female breast (110), prostate (120), cervical (120), stomach (100), lung (60), oral cavity (60), colorectal (50) [13].

An earlier estimate (2020-2022 triennium), cited in the state oncology plan, put the figure at roughly **860 new cases/year** [14] — the rise to 1,140/year may reflect a genuine upward trend (aging population, see 3.4) or improved diagnostic coverage; **the relative weight of these two factors is unknown: data not collected.**

Pediatric/adolescent cancer (Hospital da Criança e do Adolescente, the state's only pediatric reference center): 32 cases in 2022, 34 in 2023; cases requiring specialized pediatric oncology are systematically referred out of state (TFD) [14]. The state has 11 mammography units [14]. **Cancer mortality (rate, trend) and stage at diagnosis: data not collected** — the oncology plan documents only care-supply organization (radiotherapy services were still "under implementation" as of 2024) [14], with no diagnosis-stage statistics.

3.4 Possible Determinants

Epidemiological transition and aging (IBGE 2022 Census) [15] [16]: the 0-14 age share fell from 49% (1980) to 27% (2022); the 65-and-over share reached 5.5%; the aging index rose to **20.3 in 2022, from 10.4 in 2010** (nearly doubling in a decade).

Behavioral risk factors — Vigitel limitation: Vigitel covers only Macapá (the capital), not the entire state; the state's chronic-disease plan nonetheless uses this data as a proxy for the whole state, a limitation that should be flagged [9]. Vigitel-Macapá data (2019 baseline): binge alcohol consumption 16.1%, leisure-time physical activity ≥150 min/week 44.3%, ultra-processed food consumption the previous day 26.2%, smoking 7.3% [9]. Excess weight (SISVAN, 2019): 12.75% among children under 5, 29.95% among adolescents, 63.55% among adults [9].

Mercury exposure (illegal small-scale gold mining, "garimpo"): an Iepé/DSEI-AP study (192 hair samples, Karipuna, Palikur, Galibi Marworno, and Galibi Kali'na peoples, Oiapoque region) found that **50.5% of those tested had mercury levels ≥ 6.0 mg/kg** (the WHO threshold), peaking in the 51-60 age bracket (75%), with differing exposure by sex (60% of men, 38% of women, but 31.4% of women of childbearing age above the safe threshold) [17]. A Fiocruz/WWF-Brasil/IEPA/Iepé study (428 fish samples, 5 rivers, 2017-2018) found detectable mercury in 100% of samples, 28.7% above the WHO threshold, and **77.6% of the most-consumed carnivorous fish species** above that threshold; Indigenous territories affected: Wajãpi, Uaçá, Juminã, Galibi [18] [19].

Biomass-burning smoke and cardiorespiratory disease: data not collected specifically for Amapá — an equivalent study exists for Rondônia (see 4.4) but no Amapá-specific equivalent was identified in this session; this remains a plausible working hypothesis (Amapá is part of the Legal Amazon) rather than a verified, quantified finding.

Persistent infectious diseases: - *Malaria* (SIVEP-Malária, 2014-2024): **112,204 notified cases** in total (average ≈10,200/year), *P. vivax* 91.4%, Indigenous people overrepresented (8.55% of malaria cases versus 1.5% of the population), most-affected municipalities: Calçoene, Mazagão, Santana [20]. A separate source (state SVS data, confirmed cases) reports a 70% rise between 2022 (2,798 cases) and 2023 (4,726 cases), with a federal emergency declared across 7 municipalities on January 8, 2024 [21]. **Methodological note**: the two sources diverge (6,322

vs. 4,726 cases for 2023), likely due to differing definitions (notified vs. confirmed); no reconciliation was attempted. - *Dengue*: 1,082 confirmed cases through November 2023 [22]; 5,628 probable cases as of May 18, 2024 (+1,610% versus the same period in 2023), 6 confirmed deaths, ranking 6th nationally [23]. - *Tuberculosis*: two not-directly-comparable studies — 1,809 notified cases 2018-2022, 2022 incidence rate 41.4 per 100,000 (versus 35.8 per 100,000 nationally), cure rate 59.5% (WHO target: 85%) [24]; another study cites an average incidence coefficient of 71.32 per 100,000 over an unspecified period in the abstract consulted [25].

3.5 Population Profiles Affected

General demographics (IBGE 2022 Census) [15] [16]: population **733,508-733,759** (2025 estimate: 806,517), density 5.15 inhabitants/km^2, growth +9.56% (2010-2022); women 50.3%, men 49.7%.

Race/color (2022 Census) [26]: pardo 65.3% (+9.7% since 2010), white 21.4% (-2.2%), Black 11.8% (+48.7%), Indigenous 1.5% / 11,334 people (+52.9%) — Indigenous and Black populations are the two fastest-growing groups of the decade in relative terms.

Indigenous populations: the Amapá and North Pará Indigenous Health District (DSEI) is responsible for 14,470 Indigenous people across six Indigenous territories (Uaçá, Juminã, Galibi, Waiãpi, Tumucumaque Park, and a sixth not identified in the source) [9]. The Wajãpi people: roughly 2,000 people across 95 villages, with access to care requiring 2 to 5 days of travel for some isolated villages [27]. **Quantified data on Amapá's Quilombola communities: data not collected.**

Income/education (2025 PNAD Contínua) [28]: nominal monthly household income per capita R$1,697; formal employment 42.0% of those aged 14+ (2022, versus 49.1% in 2016, bases not strictly comparable); average income from main formal job R$2,998. **Precise rural/urban split and detailed education indicators: data not collected.**

3.6 Healthcare Supply Structure

Facilities (CNES, June 2025): 27 hospitals, 85.2% publicly/SUS-oriented [29].

Physician density (*Demografia Médica no Brasil 2023*, data as of June 2022) [30]: **1.84 physicians per 1,000 inhabitants**, versus **2.41 per 1,000 for Brazil**; Macapá (the capital): 2.12 per 1,000, among the lowest densities of any Brazilian state capital. **Breakdown by specialty (cardiologists, oncologists): data not collected** — a dedicated CNES-DATASUS extraction was attempted but not completed in this session.

ANS coverage (ANS TabNet, SIB/ANS/MS) [31]: 7.8% (Dec 2023), 7.8% (Dec 2024), 8.0% (Dec 2025), 7.9% (Jun 2026) — versus 25.0% for Brazil in June 2026 [4]. Amapá is one of the three lowest-coverage states in the country (ahead only of Roraima at 4.6% and Acre at 5.3%) [31]. **More than 92% of the population thus depends exclusively on SUS.**

Patient flows: the legal mechanism of Tratamento Fora de Domicílio (TFD, Ministry of Health Ordinance No. 55/1999) organizes out-of-state referrals [32]. The oncology plan confirms that pediatric oncology and (historically) radiotherapy depend on this mechanism [14]. On July 16, 2026, the Federal Public Defender's Office recommended a formal Pará-Amapá agreement to

streamline this flow toward **Belém**, following a documented case of diagnostic delay followed by death (2024) [33]. Annual oncology drug budget (2024): R$9,307,974.20 [14].

3.7 Access Inequalities

- Geographic and river-based isolation: 2 to 5 days of travel for the most isolated Wajãpi villages [27]; a continuous rise in premature deaths in the Norte micro-region (Oiapoque, Calçoene…) despite a population too dispersed for reliable rate calculation [9].
- In-hospital case-fatality: 14.5% for heart failure (2019-2024) [11].
- Diagnostic delay documented by an individual case (Public Defender's Office, 2026), but **no aggregate statistic on oncology or cardiovascular diagnostic delay: data not collected** [33].
- Very low ANS coverage (7.9%) combined with below-national physician density (1.84 vs. 2.41 per 1,000) [30] [31]: a structural bottleneck for high-complexity care.
- Unevenly distributed environmental exposure: Indigenous and riverine populations of Oiapoque disproportionately exposed to mercury (50.5% above the WHO threshold) [17] and to malaria (8.55% of cases versus 1.5% of the population) [20] [26].

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4. Rondônia (RO)

4.1 Leading Causes of Morbidity and Mortality

A direct extraction by ICD-10 chapter at the whole-state level (SIM TabNet) **could not be completed in this session** (technical limitation): **data not collected** for the overall cause-of-death structure of Rondônia. The only precise figure obtained concerns the population aged 60 and over (2001-2012, SIM/DATASUS): cardiovascular diseases accounted for **35% of deaths in this age group (12,696 deaths)**, versus 12% for respiratory diseases (4,427 deaths) [34].

Overall socio-health indicators situating the state (IBGE/UNDP aggregator) [35]: life expectancy at birth (2020) **72.2 years** (ranked 25th of 27 federative units); infant mortality (2021) **12.3 per 1,000** live births (ranked 12th); HDI (2024) **0.786** (ranked 14th); literacy (2022) **95.1%**.

A full quantified comparison (age-standardized rates) against the national and North Region averages: data not collected.

4.2 Cardiovascular and Thromboembolic Disease Burden

Reference study (Andrade Filho et al., 2017, SIM/DATASUS, ages 60+, 2001-2012) [34]: average annual age-standardized cardiovascular mortality rate among the elderly **10.0 per 1,000** (versus 3.1 per 1,000 for respiratory disease); municipality with the highest average: Ouro Preto do Oeste (14.1 per 1,000); **Porto Velho (capital): 13.6 per 1,000**; dominant causes: stroke 22%, myocardial infarction 21% of diagnoses. A complementary study (Nunes, Ignotti, Hacon, 2013) found a significant association between PM2.5 exposure from biomass burning and cardiovascular mortality among elderly people across the Brazilian Amazon in 2005, including Rondônia [36].

Hospital capacity (CNES, June 2025) [37]: **245 coronary intensive-care beds**, up sharply from 52 in 2013; 67 facilities with a dedicated cardiology service [38], including the Rondônia Heart Institute and INCOR Rondônia (Ji-Paraná). ICU beds per 10,000 inhabitants: 3.74 (2019) → 4.86 (2025), of which SUS-funded ICU beds: 0.96 → 1.49 [37].

Incidence, prevalence, and SIH hospitalizations by cardiovascular condition for Rondônia's population as a whole (all ages): data not collected (only elderly mortality could be precisely documented) — atrial fibrillation, deep vein thrombosis, and pulmonary embolism specifically: **data not collected.**

4.3 Cancer Burden

INCA estimates (*Estimativa 2023*, 2023-2025 triennium) [39] [40]:

Cancer type	Men	Women	Total/year
Breast	—	320	320
Prostate	310	—	310
Colorectal	110	100	210
Lung	110	80	190
Stomach	100	50	150
Cervical	—	150	150
Total excluding non-melanoma skin cancer	**1,200**	**1,170**	**2,370**
Total, all cancers (including skin)	**1,350**	**1,410**	**2,760**

Roughly 2,700 new diagnoses in 2023 and over 8,000 cases across the 2023-2025 triennium [40]. Most frequent types: breast (women) and prostate (men), then colorectal, lung, and stomach — the latter proportionally elevated compared to major Southern/Southeastern metropolitan areas.

Cancer mortality and stage at diagnosis for Rondônia: data not collected (would require the state's Hospital Cancer Registry, not consulted).

Reference infrastructure: **Hospital de Amor Amazônia** (Porto Velho), opened 2012, new dedicated facility in 2017, described as the largest oncology referral center in the North Region [41]; more than 2 million patient visits in 2024, 769 registered health professionals [42] [43]; expansion underway in Ji-Paraná and Vilhena (R$220 million cumulative as of July 2026) [44].

4.4 Possible Determinants

Epidemiological transition/aging: low life expectancy (72.2 years) and still relatively high infant mortality (12.3 per 1,000) suggest an incomplete transition relative to the South/Southeast [35], yet cardiovascular disease already dominates elderly mortality (35% of deaths aged 60+ in 2001-2012) [34] — a typical "double burden" coexistence.

Behavioral risk factors (Vigitel, limited to Porto Velho): per Vigitel 2021, **Porto Velho was the Brazilian capital with the highest adult overweight rate (64.41%) and obesity rate (26.41%) in 2021**, both national top ranks; 4 of the 5 capitals most affected by overweight are in the North Region [45]. National benchmark (Vigitel 2023): hypertension prevalence 27.9% (the worst result in the historical series) [46]. **Porto Velho-specific smoking/hypertension/diabetes indicators (Vigitel 2023): data not collected.**

Migration and recent territorial occupation: a textbook case of directed colonization — 2 municipalities in 1970 versus 52 today [47]; population: 36,935 (1950) → 503,125 (1980, +331.4%) → 1,130,874 (1991, +124.8%) → 1,581,196 (2022) [35], driven by the opening of highway BR-364 and INCRA's Integrated Colonization Projects in the 1970s [47]. More than 40% of the current population was born outside the state, with the main migration flows coming from the Southeast (217,424 people), the South (168,526), and the Northeast (122,335) [35]. Urbanization: 73.22% [35].

Biomass-burning smoke and cardiorespiratory disease — a **mixed** picture across studies: 112,863 fire hotspots detected 2001-2012 (average 9,405/year, peak of 17,818 in 2005), with Porto Velho accounting for 17.4% of detections [34]. Yet Andrade Filho et al. (2017) note that the spatial distribution of cardio-respiratory mortality "hotspots" among the elderly **does not directly coincide** with that of fire hotspots (long-range atmospheric transport) [34]; and Silva, Mendes, Silva (2020), covering 2009-2018, **found no direct relationship** between rising fire hotspots and rising respiratory hospitalizations in Rondônia's immediate regions [48] — a result that calls for methodological caution. At the broader Amazon scale, Nunes/Ignotti/Hacon (2013) found a significant association between biomass-burning PM2.5 and elderly cardiovascular mortality, Rondônia included [36], consistent with other work by the same research group on Amazon municipalities [49].

Mercury exposure (garimpo, Madeira River basin): a Fiocruz/ENSP technical report (2024) documents elevated contamination in fish and human populations of the Madeira basin, with results presented to the Indigenous Paiter-Suruí people [50] [51]. Rondônia was identified (2025) as a major entry point for illegal mercury feeding Amazon garimpos [52], and in August 2026 federal courts ordered the suspension of garimpo operations unable to prove the legal origin of their mercury in several states including Rondônia [53]. **The precise share of Rondônia's population above WHO thresholds: data not collected** (a study found 94% contamination, but among the Yanomami of Roraima, outside this report's scope) [54].

Persistent infectious diseases: malaria — 11,762 cases (2023) → 7,716 cases (2024, -34%) → a further 57% drop in Q1 2025 versus Q1 2024; *P. vivax* accounted for 95% of cases in 2025 [55]. The Amazon region accounts for 99% of Brazil's malaria cases [56]. **Dengue and tuberculosis (precise Rondônia figures): data not collected** — their monitoring is mentioned in the State Health Plan without figures extracted in this session [57].

4.5 Population Profiles Affected

Race/color (2022 Census) [35]: pardo 59.3%, white 30.7%, Black 8.7%, Indigenous 1.1%, Asian-descent 0.3%.

Indigenous population: 21,153 people (1.34% of the state's population, 2022 Census), sharply up from roughly 12,000 in 2010 [58]. Rondônia is the 4th-ranked Brazilian state by number of

distinct Indigenous peoples (29 peoples), with 54.54% of the Indigenous population living on Indigenous Territories [59]. The **Massaco Indigenous Territory** (421,895 ha, on the Bolivian border) hosts isolated groups, estimated at roughly 600 people in 1998; FUNAI released unprecedented footage in January 2025 of 9 isolated individuals captured in 2024; this territory is threatened by garimpo, illegal logging, and fires [59].

Urban/rural: 73.22% urbanization (~1,154,257 urban inhabitants, ~422,166 rural) [35], with riverine populations concentrated along the Madeira, Mamoré, and Guaporé rivers (Porto Velho, Guajará-Mirim), also the most exposed to mercury contamination.

Income/education: per-capita GDP (2021) R$32,044.73 [35]; monthly household income per capita (2025) R$1,991 [60]; literacy 95.1% [35]; Education-HDI historically among the lowest in the North Region (0.557 in 2010) [47].

4.6 Healthcare Supply Structure

Facilities (CNES, June 2025) [37]: 139 hospitals, of which 67 are public/SUS (48.2%); 92 general hospitals (66.2%), 16 polyclinics/specialty centers, 11 standalone day hospitals, 8 specialized hospitals. Clinical beds: 1,404 (2008) → 2,001 (2024); surgical beds 794 → 1,285; ICU beds per 10,000 inhabitants 3.74 (2019) → 4.86 (2025).

Physicians/specialists: the only figure found, old and not directly comparable to the other states' 2023 data — **11.1 physicians per 10,000 inhabitants in 2011** [47]. **A recent update (Demografia Médica 2023) and a breakdown by specialty (cardiologists, oncologists): data not collected.**

ANS coverage (ANS TabNet, Dec 2024) [5]: **Rondônia 9.1%**, versus Brazil 24.2% — regional comparison: Acre 5.0%, Roraima 4.4%, Amazonas 14.2%, Pará 10.1%, Amapá 7.8%, Tocantins 7.8%. Absolute beneficiary count: 164,857 (April 2025) versus 159,569 (April 2024) [61].

Patient flows (TFD): between 2023 and May 2024, **R$25,146,720.01** was spent on airfare for this patient transport program; the average wait to obtain a slot in another state is estimated at **roughly 30 days** [62]. **Precise destinations (Manaus, Cuiabá, Brasília, Goiânia) and volumes by condition: data not collected.**

4.7 Access Inequalities

- ANS coverage far below the national average (9.1% vs. 24.2%) [5]: near-total SUS dependence for high-complexity care.
- TFD wait time of roughly 30 days for out-of-state transfers [62], though some of the oncology need is now absorbed locally by Hospital de Amor Amazônia [42].
- Low life expectancy (72.2 years, ranked 25th of 27) and relatively high infant mortality (12.3 per 1,000) [35].
- Sharp municipal disparities in elderly cardiovascular mortality (up to 39.1 per 1,000 in Pimenteiras do Oeste in 2008) [34] and in ICU-bed distribution (Porto Velho concentrating 18.1% of the state's ICU beds) [37].
- **Premature mortality, in-hospital case-fatality by condition, and oncology diagnostic delay specific to Rondônia: data not collected.**

5. Cross-Cutting Comparative Section

5.1 Comparative Table of Key Indicators

Indicator	Tocantins	Amapá	Rondônia	Brazil / North Region
Population (2022 Census)	data not collected	733,508-733,759 [15]	1,581,196 [35]	—
Deaths, all causes (most recent year)	8,912 (2023) [6]	data not collected (aggregate)	data not collected (aggregate)	Brazil: 1,542,157 (2022) [1]
Circulatory-disease share of total mortality	28.1% (2023) [6]	data not collected (only composite NCD indicator available)	35% of deaths aged 60+ (2001-2012) [34]	Brazil: 25.8% (2022) [1]
SIH hospitalizations, circulatory disease (2023)	6,917 [7]	15,455 [7]	8,830 [7]	North Region: 80,380; Brazil: 1,279,882 [7]
Cancer — estimated cases/year (INCA, latest state estimate)	data not collected	1,140 (2026 update) [13]	2,370-2,760 (2023-2025 estimate) [39]	Brazil: 704,000 (2023-2025 triennium) [2]
ANS coverage (medical care, Dec 2024)	7.8% [5]	7.8% [5]	9.1% [5]	Brazil: 24.2% (Dec 2024) [5] / 25.0% (Jun 2026) [4]
Number of hospitals (CNES)	data not collected	27 [29]	139 [37]	—
Physician density (per 1,000 inhabitants)	data not collected	1.84 (2022) [30]	data not collected (recent); 1.11 in 2011 [47]	Brazil 2.41 [30]; North Region 1.45 [30]
Indigenous population (2022 Census)	data not collected	11,334 (1.5%) [26]	21,153 (1.34%) [58]	—
Malaria, cases/year (most recent year)	data not collected	6,322 (2023, notified) [20] or 4,726 (2023, confirmed) [21]	11,762 (2023) – 7,716 (2024) [55]	Amazon region: ~99% of Brazil's cases [56]

5.2 Cross-Cutting Reading

Supplementary health coverage. All three states show ANS coverage far below the national average, and the North Region is consistently described in secondary literature as the country's lowest-coverage region [63] [64], although the precise aggregated regional coverage figures (the North Region as a whole) were not cross-checked directly on the ANS portal in this session (flagged "not verified against the primary source" by the agent that reported them) [63] [64]. This finding is **established by primary state-level data** (ANS TabNet) for all three states studied, while its generalization to the region as a whole rests on less-verified secondary sources.

Cancer. Based on the literature and INCA benchmarks consulted, cervical cancer is the most incident cancer in the North Region (unlike the national level, where it ranks only 4th among women), notably in Amapá (adjusted rate 31.13 per 100,000 women per the 2026-2028 estimate) — a structural determinant cited in the literature is historically the lowest HPV vaccination coverage in the country, found in the Northern states [65] [66]. Stomach cancer is also cited as proportionally elevated in the North Region [67], consistent with the figures obtained for Amapá and Rondônia (100 and 150 cases/year respectively). INCA also projects the North Region will see the country's steepest rise in premature prostate-cancer mortality (+52%, versus +37% in the Northeast) [68].

Specialist density. Per the 2023 SBOC/Datafolha census, the North Region accounts for only **3% of the country's clinical oncologists** [69] [70], and per *Demografia Médica no Brasil 2023*, overall physician density there is Brazil's lowest (1.45 per 1,000 versus a 2.60 per 1,000 national figure per the version consulted by the cross-cutting research agent — a slight discrepancy with the 2.41 per 1,000 figure cited by the Amapá-focused agent exists between the two extractions of the same report, to be reconciled by direct consultation of the source document) [30] [30]. No precise regionalized data on the number of cardiologists could be obtained for any of the three states: **data not collected**, uniformly.

Shared environmental determinants. Mercury exposure linked to illegal small-scale gold mining is documented by independent studies in Amapá [17] [18] and Rondônia [50] [52], within a national governance framework that is tightening (a 2025-2030 ministerial strategic plan [71], federal court rulings in 2026 [53]). The link between biomass-burning smoke and cardiorespiratory morbidity, however, **is not confirmed unambiguously in the literature reviewed**: a regional study (Nunes/Ignotti/Hacon 2013) finds a significant association at the Amazon-wide scale [36] [49], while a Rondônia-specific study (Silva/Mendes/Silva 2020) finds no direct correlation over 2009-2018 [48] — this point should be treated as a contested hypothesis rather than an established fact.

Residual infectious diseases. Malaria remains a significant burden in both Amapá and Rondônia (both with thousands of annual cases), with recent improvement in Rondônia (-34% then -57%) [55] but a resurgence in Amapá in 2023-2024 [20] [21]. The coexistence of this infectious burden with the rise of non-communicable chronic disease is documented conceptually in the literature on the "double/triple burden of disease" in the Amazon [72] [73] [74], though no single combined figure was found for any of the three states.

Patient flows. The TFD mechanism exists and is active in all three states [32] [62] [8], but **no verifiable quantified data (volumes, precise destinations, comparative wait times) could be obtained for Tocantins or Amapá**; only Rondônia has an aggregate cost figure (R$25.1 million in airfare, 2023-May 2024) and an average wait time (~30 days) [62]. This is a literature/open-data gap identified by this research, and potentially an original contribution angle for the ShokAI project via analysis of the three states' SISREG (patient-referral regulation systems).

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6. Testable Hypotheses on ANS/TISS Data

These three hypotheses link a determinant documented above to an expected "patient potential," and are formulated to be directly testable on ANS/TISS microdata (available columns: operator, provider, procedure code, event value, age, sex, municipality).

H1 — Geographic concentration of insured patient potential. Given the very low ANS coverage across the three states (7.8-9.1% versus roughly 24-25% nationally) [5] [4], the volume and value of TISS cardiology and oncology events should be heavily concentrated in each state's capital municipality (Palmas, Macapá, Porto Velho) and its immediate metropolitan area, at a share well above that municipality's demographic weight within the state. *Proposed test*: aggregate TISS event counts and values by provider municipality, compute the ratio (share of events / share of IBGE population) for the capital versus the rest of the state, and test whether this ratio is significantly greater than 1.

H2 — Cardiovascular over-incidence in garimpo-affected municipalities. Given documented elevated mercury exposure and an associated cardiovascular risk in municipalities known for small-scale illegal gold-mining activity (e.g., Oiapoque and Calçoene in Amapá [20] [17]; the Madeira River basin in Rondônia [50] [52]), health-plan beneficiaries residing in these municipalities should show a higher rate of cardiovascular procedure-code use (hypertension follow-up, arrhythmia, stroke-related codes) per beneficiary than in comparably-sized municipalities unaffected by garimpo. *Proposed test*: cross-reference TISS procedure codes by municipality against a list of municipalities known for garimpo activity (drawn from the sources cited in 3.4/4.4), normalize by beneficiary count (age/sex-adjusted), and compare the two groups of municipalities.

H3 — Aging of the cardiovascular patient potential. Given the documented rise in Amapá's aging index (10.4 in 2010 → 20.3 in 2022) [15] [16] and the rise in premature non-communicable-disease mortality over the same period (214.1 → 252.0 per 100,000) [9], the share of total TISS cardiovascular event value attributable to the oldest age brackets (60 and over) should rise year over year for a given operator/municipality. *Proposed test*: track, for each provider municipality and each operator, the year-by-year evolution of the share of cardiovascular event value (relevant TUSS codes) attributable to beneficiaries aged 60 and over, and test the trend for statistical significance.

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7. References (Vancouver)

1. D'Alegria B. Atualização – Dados do DATASUS 2022. Estratégia MED [Internet]. 25 out 2023. Available from: https://med.estrategia.com/portal/atualidades/atualizacao-dados-datasus-2022/
2. INCA estima 704 mil casos de câncer por ano no Brasil até 2025. Portal Gov.br/INCA [Internet]. 23 Nov 2022.
3. Instituto Nacional de Câncer (INCA). Estimativa 2023: incidência de câncer no Brasil. Rio de Janeiro: INCA; 2022. Available from: https://www.inca.gov.br/publicacoes/livros/estimativa-2023-incidencia-de-cancer-no-brasil
4. Agência Nacional de Saúde Suplementar (ANS). Dados Gerais — Perfil do Setor [Internet]. Data as of June 2026. Available from: https://www.gov.br/ans/pt-br/acesso-a-informacao/perfil-do-setor/dados-gerais
5. Agência Nacional de Saúde Suplementar (ANS). TabNet ANS — Taxa de Cobertura de Planos de Saúde, Assistência Médica segundo UF, Dec 2024 [Internet]. Available from: https://www.ans.gov.br/anstabnet/cgi-bin/dh?dados/tabnet_tx.def
6. Ministério da Saúde/SVSA/CGIAE. Sistema de Informações sobre Mortalidade (SIM) — Óbitos por residência segundo Capítulo CID-10, Tocantins, 2023 [TabNet, accessed 24 Aug 2026]. Available from: https://tabnet.datasus.gov.br/cgi/tabcgi.exe?sim/cnv/obt10to.def
7. Ministério da Saúde/DATASUS. Sistema de Informações Hospitalares do SUS (SIH/SUS) – Morbidade hospitalar do SUS por local de internação, Capítulo CID-10 IX (doenças do aparelho circulatório), Brasil, 2023 [TabNet, accessed 24 Aug 2026]. Available from: http://tabnet.datasus.gov.br/cgi/tabcgi.exe?sih/cnv/niuf.def
8. Governo do Tocantins. TFD Palmas-TO. Portal de Serviços [Internet]. Available from: https://central3.to.gov.br
9. Governo do Estado do Amapá. Secretaria de Estado da Saúde (SESA), Superintendência de Vigilância em Saúde (SVS). Plano Estadual de Ações Estratégicas para Enfrentamento de Doenças Crônicas e Agravos Não Transmissíveis (DANT) do Amapá 2024-2030. Macapá: SESA; 2024. Available from: https://editor.amapa.gov.br/arquivos_portais/livraria/SESA_163af87086ebf49974b8debfa1ad5cea.pdf
10. Pantoja CSM, Dantas JO, Santos DN, Sfair MM, Klein CO, Barbosa IC, et al. Mortalidade por Acidente Vascular Cerebral e Infarto Agudo do Miocárdio no estado do Amapá, 2015-2024. Rev Eletrônica Acervo Saúde. 2025;25(6):e20706. doi:10.25248/reas.e20706.2025.
11. Damasceno ÁAM. Insuficiência cardíaca no Amapá: análise de tendência de cinco anos. Rev Delos. 2025;18(63):e3634. doi:10.55905/rdelosv18.n63-113.
12. Instituto Nacional de Cardiologia (INC)/Ministério da Saúde. Painel Dashcardio — Mortalidade cardiovascular na Amazônia Legal [Internet]. Available from: https://inc.saude.gov.br/htm/dashcardio.htm
13. Instituto Nacional de Câncer (INCA). Amapá — estimativa dos casos novos [Internet]. Rio de Janeiro: INCA/Ministério da Saúde; updated 04 Feb 2026. Available from: https://www.gov.br/inca/pt-br/assuntos/cancer/numeros/estimativa/estado-capital/amapa

14. Governo do Estado do Amapá. SESA. 1º Plano Estadual de Assistência Oncológica do Amapá. Macapá: SESA; 2024. Available from: https://editor.amapa.gov.br/arquivos_portais/livraria/SESA_a4b059933cb842fb6a9960bfab6d5710.pdf
15. Instituto Brasileiro de Geografia e Estatística (IBGE). Cidades@ | Amapá | Panorama [Internet]. Available from: https://cidades.ibge.gov.br/brasil/ap/panorama
16. Diário do Amapá. Censo 2022: IBGE divulga dados sobre idade e sexo da população do Amapá. 28 Oct 2023.
17. G1 Amapá. Estudo revela contaminação por mercúrio em indígenas no Amapá [Iepé/DSEI-AP study]. 3 Jun 2026.
18. Bartaburu X. Peixes mais consumidos no Amapá estão contaminados pelo mercúrio do garimpo. Mongabay Brasil. Aug 2020.
19. Hacon SS, Oliveira-da-Costa M, Gama CD, Ferreira R, Basta PC, et al. Mercury Exposure through Fish Consumption in Traditional Communities in the Brazilian Northern Amazon. Int J Environ Res Public Health. 2020;17(15):5269. doi:10.3390/ijerph17155269.
20. Oliveira AG, et al. Malária no Amapá 2014-2024: Análise epidemiológica dos casos notificados. Rev Cient Multidiscip Núcleo Conhecimento. 2025;1(8):19-38.
21. Seles na Fes. Malária: após aumento de 70%, União reconhece emergência em 7 cidades do AP. Jan 2024.
22. Diário do Amapá. Até novembro de 2023, Amapá registrou mais de mil casos de dengue. 5 Dec 2023.
23. G1 Amapá. Amapá apresenta 'tendência de queda' nos casos de dengue, segundo Ministério da Saúde. 21 May 2024.
24. Cardoso RF, Braga FAG, Marçal PHEC, Abreu STQ, Carvalho DM, Silva Junior JP, et al. Aspectos clínicos e epidemiológicos da tuberculose no estado do Amapá. Braz J Health Rev. 2023;6(3):12689-12703. doi:10.34119/bjhrv6n3-333.
25. [Authorship unconfirmed]. Aspectos clínicos e epidemiológicos da tuberculose no estado do Amapá [distinct study, abstract only]. Available from: https://www.researchgate.net/publication/372054403
26. G1 Amapá. Censo 2022: população indígena e preta foram as que mais cresceram no Amapá. 22 Dec 2023.
27. O Liberal. Waiãpi sob ameaça: garimpo e escassez de saúde trazem riscos ao povo da floresta. 5 Aug 2025.
28. Instituto Brasileiro de Geografia e Estatística (IBGE). Cidades@ | Amapá | Panorama — Trabalho e Rendimento (PNAD Contínua 2025) [Internet]. Available from: https://cidades.ibge.gov.br/brasil/ap/panorama
29. Fundação Oswaldo Cruz (Fiocruz). Cadastro Nacional de Estabelecimentos de Saúde (CNES) — Painel Amapá, June 2025 [Internet]. Available from: https://www.tabnet.fiocruz.br/dash/dash_ap01.html
30. Scheffer M, et al. Demografia Médica no Brasil 2023. São Paulo: FMUSP/AMB; 2023. ISBN 978-65-00-60986-8. Available from: https://amb.org.br/lancada-a-demografia-medica-no-brasil-2023/

31. Agência Nacional de Saúde Suplementar (ANS). TabNet — Taxa de Cobertura de Planos de Saúde, Assistência Médica segundo UF (source SIB/ANS/MS, June 2026) [Internet]. Available from: https://www.ans.gov.br/anstabnet/cgi-bin/dh?dados/tabnet_tx.def
32. Wiki Regulação/Ministério da Saúde. Tratamento Fora do Domicílio (TFD). Portaria MS nº 55/1999 [Internet]. Available from: https://wiki.saude.gov.br/regulacao/index.php/Tratamento_Fora_do_Domicílio_(TFD)
33. Defensoria Pública da União (DPU). DPU recomenda que governos do Pará e Amapá realizem convênio para tratamento de pacientes. 16 Jul 2026.
34. Andrade Filho VS, et al. Distribuição espacial de queimadas e mortalidade em idosos em região da Amazônia Brasileira, 2001-2012. Ciênc Saúde Colet. 2017;22(1):245-253.
35. Wikipédia. Rondônia [Internet, synthesizing IBGE 2022 Census/Panorama, SIM, PNUD Radar IDHM data]. Available from: https://pt.wikipedia.org/wiki/Rondônia
36. Nunes KVR, Ignotti E, Hacon SS. Circulatory disease mortality rates in the elderly and exposure to PM2.5 generated by biomass burning in the Brazilian Amazon in 2005. Cad Saúde Pública. 2013;29(3):589-598.
37. Fundação Oswaldo Cruz (Fiocruz). Painel Cadastro Nacional de Estabelecimentos de Saúde (CNES) — Rondônia [Internet]. Available from: https://tabnet.fiocruz.br/dash/dash_ro01.html
38. DATASUS/Ministério da Saúde. CNES — Indicadores de Serviços Especializados: Serviço de Atenção Cardiovascular/Cardiologia, Estado de Rondônia [Internet]. Available from: https://cnes2.datasus.gov.br/Mod_Ind_Especialidades_Listar.asp?VTipo=116&VListar=1&VEstado=11
39. Instituto Nacional de Câncer (INCA). Rondônia – estimativa dos casos novos [PDF]. Available from: https://ideiasus.fiocruz.br/wp-content/uploads/2025/04/2-Rondonia-estimativa-dos-casos-novos-Instituto-Nacional-de-Cancer-INCA.pdf
40. Ministério da Saúde/Fiocruz IdeiaSUS. Estudo aponta para estimativa de mais de 8 mil novos casos de câncer em Rondônia até 2025 [PDF]. Available from: https://ideiasus.fiocruz.br/wp-content/uploads/2025/04/1-Estudo-aponta-para-estimativa-de-mais-de-8-mil-novos-casos-de-cancer-em-Rondonia-—-Ministerio-da-Saude_.pdf
41. Hospital de Amor. Hospital de Amor Amazônia [Internet]. Available from: https://hospitaldeamor.com.br/hospital-de-amor-amazonia/
42. InformaRondônia. Sílvia Cristina destaca atuação do Hospital de Amor com mais de 2 milhões de atendimentos em 2024. 25 Mar 2025.
43. Guia Saúde Brasil. Hospital de Amor Amazônia — profissionais [Internet]. Available from: https://guiasaudebrasil.com.br/unidade/rondonia/porto-velho/7068336-hospital-de-amor-amazonia/profissionais
44. InformaRondônia. Sílvia Cristina destina mais R$42 milhões ao Hospital de Amor e acumula R$220 milhões para oncologia em Rondônia. 3 Jul 2026.
45. G1 Rondônia. Porto Velho é capital com maior porcentagem de adultos com excesso de peso e obesidade [Vigitel 2021 data]. 12 Apr 2022.
46. Instituto Nacional de Cardiologia (INC)/Ministério da Saúde. A Hipertensão avançou no Brasil nos últimos anos [Vigitel 2023 data] [Internet]. Available from: https://inc.saude.gov.br/htm/noticia85.htm

47. Santos TRS, Ricarte CAA, Conceição FSD. Rondônia – mutações de um território federal na Amazônia brasileira. Confins [Online]. 2016;(29). doi:10.4000/confins.11569.
48. Silva LB, Mendes DFS, Silva RMP. Impactos da poluição das queimadas à saúde humana: internações por doenças respiratórias no Estado de Rondônia entre 2009 e 2018. ID on line Rev Psicol. 2020;14(52):414-427. doi:10.14295/idonline.v14i52.2713.
49. Ignotti E, Hacon SS, Junger WL, et al. Efeitos das queimadas na Amazônia: método de seleção dos municípios segundo indicadores de saúde. Rev Bras Epidemiol. 2007;10(4):453-464.
50. Basta PC, Vasconcellos AC, et al. Impacto do mercúrio em áreas protegidas e povos da floresta na Amazônia: uma abordagem integrada saúde-ambiente. Relatório Técnico. Rio de Janeiro: Fiocruz/ENSP; 2024. 173p. Available from: https://arca.fiocruz.br/items/0099696d-8092-4a03-afbb-9d2b837433cb
51. Fiotec/Fiocruz. Fiocruz apresenta resultados do estudo 'Impacto do mercúrio em áreas protegidas e povos da Amazônia' [Internet].
52. G1 Rondônia. Como Rondônia se tornou a nova porta de entrada do mercúrio ilegal que abastece garimpos na Amazônia. 15 Nov 2025.
53. G1 Rondônia. Justiça Federal ordena suspensão de garimpos que não provarem origem lícita de mercúrio na Amazônia. 23 Aug 2026.
54. Um Só Planeta/G1. Exames revelam presença de mercúrio em amostras de cabelo de yanomamis. 4 Apr 2024.
55. Governo do Estado de Rondônia/Agevisa. Boletim aponta que Rondônia apresenta queda de 57% nos casos de malária em 2025. 10 Apr 2025.
56. G1 Rondônia. Rondônia registra mais de 2 mil casos de malária nos primeiros meses de 2023. 28 Apr 2023.
57. Governo do Estado de Rondônia/Secretaria de Estado da Saúde (SESAU). Plano Estadual de Saúde 2024/2027. Porto Velho: SESAU; 2024.
58. G1. Censo do IBGE: Rondônia tem mais de 21 mil indígenas. 7 Aug 2023.
59. Wikipédia. Terra Indígena Massaco [Internet]. Available from: https://pt.wikipedia.org/wiki/Terra_Indígena_Massaco
60. Instituto Brasileiro de Geografia e Estatística (IBGE). Rondônia | Cidades e Estados [Internet]. Available from: https://www.ibge.gov.br/cidades-e-estados/ro.html
61. Agência Nacional de Saúde Suplementar (ANS). Dados de beneficiários por UF, abril de 2025 [PDF]. Available from: https://static.poder360.com.br/2025/06/ANS-beneficiarios-abril2025.pdf
62. Governo do Estado de Rondônia. Atendimento para Tratamento Fora de Domicílio é realizado na Secretaria Estadual de Saúde. 14 Jun 2024.
63. Apenas 13 a cada 100 habitantes do Nordeste têm plano de saúde. Agência Tatu / Tribuna União. 1 Mar 2024.
64. Cresce adesão a planos de saúde no AM. Amazonas Atual. 19 May 2025.
65. Associação Médica Brasileira (AMB). Câncer de colo de útero é mais frequente no Norte; Sul e Sudeste concentram casos gástricos [Internet]. Available from: https://amb.org.br/cancer-de-colo-de-utero-e-mais-frequente-no-norte-sul-e-sudeste-concentram-casos-gastricos/

66. Martins LFL, et al. Estimativas para o Triênio 2026-2028: Incidência de Câncer no Brasil. Rev Bras Cancerol / SciELO Brasil. 2026.
67. O padrão da mortalidade por câncer nas regiões brasileiras. Observatório de Oncologia [Internet].
68. INCA prevê aumento da mortalidade prematura por câncer de próstata nas regiões brasileiras. Portal Gov.br/INCA [Internet]. 27 Jan 2023.
69. Sociedade Brasileira de Oncologia Clínica (SBOC). São Paulo tem 1.763 oncologistas clínicos. SBOC/Datafolha (Censo SBOC da Oncologia Clínica 2023). 10 Dec 2025.
70. Maioria dos oncologistas do Brasil são brancos e atuam na região Sudeste. Abrale. 29 Nov 2023.
71. Ministério da Saúde. Plano Estratégico para Medidas de Atenção, Vigilância e Promoção à Saúde de Populações Expostas ao Mercúrio (2025-2030).
72. Soares Filho AM, et al. Tripla carga de doenças no Brasil, 1990-2021: mudanças temporais segundo o Estudo Carga Global de Doença. Universidade Federal de Minas Gerais; 2022.
73. Borges WD. [Master's thesis on epidemiological transition and non-communicable chronic diseases in Amazonas]. Manaus: Universidade Federal do Amazonas (UFAM); 2011.
74. Confalonieri UE. Saúde na Amazônia: um modelo conceitual para a análise de paisagens e doenças. Estudos Avançados. 2005;19(53).